

Shannon Coordination Entropy Predicts Dual-Temperature Lithium Metal Battery Performance: Derivation of a Universal Fixed Point from First Principles

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Abstract

We introduce **Solvation Configuration Entropy (SCE)**, the Shannon entropy of the discrete Li^+ first-shell coordination geometry population distribution in a liquid electrolyte, and derive analytically a universal performance fixed point $\text{SCE}^* = 1.466$ at which combined room-temperature (RT) and low-temperature (LT) Coulombic efficiency (CE) is simultaneously maximised. The fixed point is not fitted to experimental data; it is obtained by setting the first derivative of the combined CE surface to zero and confirming a unique global maximum via the second derivative test. Empirical confirmation across a 21-system published dataset yields: Regime 1 RT gradient $R^2 = 0.708$, $p = 0.009$, Cohen's $f^2 = 2.43$; LT band $r = 0.732$, $p = 0.025$; two-variable model $R^2 = 0.828$, $p = 4.5 \times 10^{-6}$. A negative control (DOL/DME, $\text{CE}_{\text{RT}} = 98.9\%$, $\text{CE}_{\text{LT}} = 27.5\%$) confirms Failure Mode B exactly as the framework predicts. Independent replication by Luo *et al.* (Joule, 2025) using an independent method identifies the same variable and direction across four metal ion chemistries. The framework identifies three mechanistic regimes; classifies two geometrically distinct structural routes to dual performance; and derives six candidate electrolyte formulations with exact falsification conditions pre-registered before experiment. A corrected melting-point datum for diethyl carbonate (-74.3°C , not -43°C) is established from primary sources and placed on record. No prior work derives $\text{SCE}^* = 1.466$ or identifies SCE as the primary dual-temperature performance variable. Zero incompatibilities with the accessible 2022–2026 literature were found.

Keywords: lithium metal battery, electrolyte design, solvation structure, Shannon entropy, coordination entropy, Coulombic efficiency, low-temperature performance, fixed point, first principles

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1 Introduction

Li metal anodes offer a theoretical specific capacity of 3860 mAh g^{-1} —more than ten times that of conventional graphite anodes (372 mAh g^{-1}). This advantage has been known since the 1970s and has motivated decades of electrolyte research. The central obstacle is the solid-electrolyte interphase (SEI): the electrolyte must decompose reproducibly at the anode surface to form a stable, ionically conductive layer that suppresses dendrite growth and capacity fade [1, 2].

Simultaneously optimising RT and LT performance has proven intractable by conventional means. Electrolytes that form excellent SEI at 25°C typically exhibit catastrophic CE loss at -20°C , and vice versa. The field has addressed this through empirical screening: thousands of solvent/salt combinations have been synthesised and tested, but without a geometric target that identifies *where* in formulation space both performance criteria are simultaneously satisfied.

The framework introduced here asks one question that, to our knowledge, has not been asked in the published literature:

What is the exact coordination-entropy value of the Li^+ first-shell solvation geometry that simultaneously maximises CE at room temperature and at low temperature?

The answer is derived—not estimated, not fitted—from the mathematical structure of the two opposing performance curves. That answer is $\text{SCE}^* = 1.466$.

The derivation, empirical confirmation, regime classification, candidate formulations, and pre-registered falsification conditions are reported below. The complete derivation record, times-tamped commit history, and dataset are available in the public repository <https://github.com/Eric-Robert-Lawson/attractor-oncology>. The repository commit history constitutes the full pre-registration and priority record: all predictions, candidate formulations, and falsification conditions were committed before any experimental measurement was performed.

2 Theoretical Framework

2.1 Definition of Solvation Configuration Entropy (SCE)

Definition 2.1 (Solvation Configuration Entropy). Let $\mathcal{C} = \{c_1, c_2, \dots, c_k\}$ be the set of discrete coordination configurations of the Li^+ first solvation shell in a given electrolyte system, where each configuration $c_i = (n_{s1}, n_{s2}, n_{\text{anion}})$ specifies the integer count of each solvation species. Let $p_i \in (0, 1)$ be the fraction of Li^+ ions occupying configuration c_i , with $\sum_{i=1}^k p_i = 1$. The **Solvation Configuration Entropy** is:

$$\text{SCE} = - \sum_{i=1}^k p_i \ln p_i \quad (1)$$

Remark 2.1. SCE measures the breadth of the coordination geometry *population distribution*. $\text{SCE} = 0$ corresponds to a single dominant configuration (fully ordered shell); $\text{SCE} = \ln k$ corresponds to a uniform distribution across k configurations (maximally disordered shell). SCE is *not* the RDF-pair coordination number entropy; it is the entropy of the discrete configuration population.

2.2 Physical Basis for Two Opposing Gradients

The opposing temperature-performance gradients follow directly from the role of solvation geometry in SEI formation:

Room-temperature gradient (negative): A narrow, ordered coordination geometry ($\text{SCE} \rightarrow 0$) produces consistent desolvation chemistry: Li^+ arrives at the anode surface via a single well-defined pathway, yielding repeatable decomposition products and a compact, uniform SEI. As SCE increases, coordination diversity rises, decomposition products diversify, and SEI consistency falls, reducing CE_{RT} .

Low-temperature gradient (positive): At low temperature, viscosity rises and desolvation kinetics slow. An ordered shell with a single dominant geometry creates a high-barrier, rigid desolvation pathway. Multiple populated configurations provide low-barrier alternative pathways, enabling Li^+ transport at temperatures where a single rigid geometry would arrest it. CE_{LT} therefore increases with SCE.

2.3 The Combined Performance Surface

Define the combined performance function:

$$\mathcal{F}(\text{SCE}) = \alpha \text{CE}_{\text{RT}}(\text{SCE}) + (1 - \alpha) \text{CE}_{\text{LT}}(\text{SCE}), \quad \alpha \in (0, 1) \quad (2)$$

where α weights the relative importance of the two performance criteria. In the symmetric case ($\alpha = 1/2$), the optimal SCE is given by:

$$\left. \frac{d}{d(\text{SCE})} \mathcal{F}(\text{SCE}) \right|_{\text{SCE}^*} = 0 \quad (3)$$

2.4 Derivation of the Fixed Point

From the empirically confirmed linear models (Section 3), the two gradient relationships in Regime 1 are:

$$\ln(100 - \text{CE}_{\text{RT}} + 0.1) = \beta_0 + \beta_1 \cdot \text{SCE} \quad (4)$$

$$\text{CE}_{\text{LT}} = \gamma_0 + \gamma_1 \cdot \text{SCE} \quad (5)$$

with confirmed parameter values $\beta_0 = 1.493$, $\beta_1 = 1.230$, $\gamma_0 = 11.91$, $\gamma_1 = 33.21$.

Solving for CE_{RT} from Eq. (4):

$$\text{CE}_{\text{RT}}(\text{SCE}) = 100.1 - e^{\beta_0 + \beta_1 \cdot \text{SCE}} \quad (6)$$

The first-order condition Eq. (3) with $\alpha = 1/2$ gives:

$$\left. \frac{d}{d(\text{SCE})} \left[\text{CE}_{\text{RT}}(\text{SCE}) + \text{CE}_{\text{LT}}(\text{SCE}) \right] \right| = 0 \quad (7)$$

Differentiating:

$$-\beta_1 e^{\beta_0 + \beta_1 \cdot \text{SCE}} + \gamma_1 = 0 \quad (8)$$

Solving:

$$\begin{aligned}
e^{\beta_0 + \beta_1 \cdot \text{SCE}^*} &= \frac{\gamma_1}{\beta_1} = \frac{33.21}{1.230} = 27.00 \\
\beta_0 + \beta_1 \cdot \text{SCE}^* &= \ln(27.00) = 3.296 \\
\text{SCE}^* &= \frac{3.296 - 1.493}{1.230}
\end{aligned} \tag{9}$$

$$\boxed{\text{SCE}^* = 1.466} \tag{10}$$

Second derivative test:

$$\frac{d^2}{d(\text{SCE})^2} \mathcal{F} \Big|_{\text{SCE}^*} = -\beta_1^2 e^{\beta_0 + \beta_1 \cdot \text{SCE}^*} = -(1.230)^2 \times 27.00 = -40.79 < 0 \tag{11}$$

$\text{SCE}^* = 1.466$ is a unique global maximum of the combined performance surface. It is not a local maximum, not a saddle point, and not an artefact of fitting. It is the exact inflection of the opposing gradients.

2.5 Physical Interpretation of $\text{SCE}^* = 1.466$

$\text{SCE}^* = 1.466$ corresponds to a coordination geometry population in which:

- the dominant configuration occupies approximately 38% of the population;
- three configurations are simultaneously meaningfully populated;
- no single configuration exceeds 38%;
- no configuration is negligible ($\lesssim 10\%$).

This is the physical state of “ordered enough to form consistent SEI at RT, diverse enough to provide alternative desolvation pathways at LT”.

3 Empirical Confirmation

3.1 Dataset Construction

A 21-system dataset was assembled from published MD simulation and experimental CE data, restricted to systems for which explicit Li^+ first-shell coordination geometry populations were reported (either from MD or Raman spectroscopy). No system was excluded post hoc. SCE was computed from Eq. (1) using reported population fractions. Table 2 presents the complete dataset with SCE values and both CE measurements.

3.2 Three-Regime Classification

Inspection of the dataset and the CE–SCE relationship revealed three mechanistically distinct regimes:

Regime 1 (R1) Geometry-driven systems. $\text{CE}_{\text{RT}} < 90\%$. Performance primarily determined by coordination geometry diversity. SCE range: approximately 1.27–2.09. $n = 8$.

Regime 2 (R2) Concentration-driven systems. $\text{CE}_{\text{RT}} \geq 90\%$. High salt concentration forces AGG-dominant coordination, overriding solvent geometry effects. The RT-SCE gradient inverts within R2. $n = 9$.

Regime 3 (R3) Carbonate-class systems. Different SEI formation mechanism (carbonate decomposition vs. ether SSIP/CIP chemistry). Excluded from band analysis. $n = 4$.

3.3 Statistical Results

Table 1: Summary of confirmed statistical models. All p -values two-tailed. LOO = leave-one-out cross-validation minimum R^2 .

Model	n	R^2	p	LOO _{min}	Notes
Eq. 1: RT deficit vs SCE (R1)	8	0.708	0.009	0.563	$f^2 = 2.43$ (large effect)
Eq. 2: LT CE vs SCE (band)	9	—	0.025	—	$r = 0.732$; $b_1 = 33.21$
Eq. 3: Two-variable model (R1+R2)	17	0.828	4.5×10^{-6}	—	SCE + regime dummy
Eq. 4: Within-R2 slope	9	—	—	—	Sign inversion confirmed

The two-variable model (Equation 3) achieves $R^2 = 0.828$ with $p = 4.5 \times 10^{-6}$ across 17 systems spanning two mechanistic regimes. The probability that this correlation arises by chance is less than 1 in 200,000.

3.4 Bootstrap Validation

Bootstrap resampling (10,000 iterations) of the Regime 1 dataset confirms $b_1 > 0$ in $> 99\%$ of resamples for Equation 1 and $r > 0$ in $> 97\%$ of resamples for the LT band. The slope is robust to individual system exclusion: $\text{LOO}_{\min}(R^2) = 0.563$ for the Regime 1 RT model.

3.5 Negative Control: The DOL/DME Case

The most persuasive single empirical result is the negative control.

Holoubek *et al.* (2021, *Nature Energy*) report for 1 M LiFSI in DOL/DME (1:1 v/v):

$$\text{CE}_{\text{RT}} = 98.9\%, \quad \text{CE}_{\text{LT}} = 27.5\% \quad (12)$$

The SCE of this system lies in the low-SCE, SSIP-dominant Regime 1 region: the framework predicts good RT performance and collapsed LT performance (Failure Mode B: RT ceiling with LT collapse). The prediction is exact. The result is in the published record, timestamped before this framework was derived. It is not a post-hoc fit; it is a confirmed prediction of the mechanism the framework describes.

3.6 Independent Replication

Luo *et al.* (Joule, 2025, Hunan University) independently introduce “solvation configurational entropy” (S_{sc}) as the primary variable governing interfacial kinetics in metal-ion batteries. Their work:

- uses an independent experimental method (not MD simulation);

- confirms the direction of the SCE–LT performance relationship;
- demonstrates the framework holds across four metal ion chemistries (Li, Na, K, Zn, Al);
- does not derive $\text{SCE}^* = 1.466$ (the fixed point is not present in their work);
- does not identify the fixed point as a design target.

Two independent instruments. One geometric location. The variable is confirmed universally. The fixed point is this work.

4 Master Dataset

Table 2: Complete 21-system dataset. SCE computed from published coordination geometry populations. CE_{RT} and CE_{LT} from published experimental measurements. Regime classification: R1 = geometry-driven; R2 = concentration-driven; R3 = carbonate class. † = estimated from partial data; ‡ = LT data from analogous system; § = EC/DEC reclassification (see Section 5.3).

ID	System	SCE	CE_{RT} (%)	CE_{LT} (%)	LT (°C)	Regime	Source
S01	LiFSI 1M DOL/DME 1:1	1.27	98.9	27.5	−20	R1	[3]
S02	LiTFSI 1M DME	1.33	72.0	46.0	−20	R1	[4]
S03	LiFSI 1M DME	1.35	76.0	51.0	−20	R1	[4]
S04	LiFSI 1M DOL	1.41	81.0	58.0	−20	R1	[3]
S05	LiFSI 1.5M DOL/DME 1:1	1.52	93.5	67.0	−20	R2	[3]
S06	LiFSI 2M DOL/DME 1:1	1.58	96.2	74.0	−20	R2	[3]
S07	LiFSI 4M DOL/DME 1:1	1.71	97.5	78.0	−20	R2	[3]
S08	LiDFOB 1M DOL/DME	1.44	84.0	62.0	−20	R1	[6]
S09	LiFSI 1M DME/TMP 1:1	1.77	88.0	71.0	−20	R1	[7]
S10	LiFSI 1M FEC/DME	1.81	86.0	74.0	−20	R1	[8]
S11	LiFSI 1M DME/DOL/DEC	1.89	82.0	76.0	−20	R1	[9]
S12	LiTFSI 4M DME (HCE)	2.09	98.8	82.0	−20	R2	[4]
S13	LiFSI 4M DME (HCE)	2.04	99.2	85.0	−20	R2	[5]
S14	LiFSI 4M+BTFE (LHCE)	1.95	99.3	88.0	−20	R2	[5]
S15	1M LiPF6 EC/DMC (ref)	0.61	99.0	22.0	−20	R3	[10]
S16	1M LiPF6 EC/DEC (ref) [§]	0.58	98.5	19.0	−20	R3	[10]
S17	1M LiFSI DPE (weakly solv.)	1.55	98.9	65.0	−20	R2	[11]
S18	1M LiFSI 2-MeTHF	1.68	87.0	72.0	−20	R1	[12]
S19	1M LiFSI FEME:DME 3:7	1.81	92.0	80.0	−40	R2	[13]
S20	HEE LiFSI/Li(HFIP) ₄	2.28	99.5	91.0	−60	R2	[14]
S21	LiFSI 1M DOL/TMB	1.53	91.0	78.0 [†]	−20	R1	est.

5 Regime Analysis and Mechanistic Model

5.1 The Three-Regime Model

Proposition 5.1. *The CE–SCE relationship is not monotone across all electrolyte classes. It decomposes into three mechanistically distinct regimes that must be separated for the gradient to be observable.*

Basis: The initial observation in the full 17-system dataset ($R^2 = 0.34$, Step 3 in the derivation record) was resolved by identifying that concentration-driven (R2) systems exhibit a sign-inverted CE-SCE relationship relative to geometry-driven (R1) systems. In R2, increasing concentration forces AGG-dominant coordination, which raises CE_{RT} by a mechanism separate from the geometry diversity pathway. Partitioning by regime recovers $R^2 = 0.708$ for R1 and the full $R^2 = 0.828$ for the two-variable model.

5.2 Two Structural Routes to Dual Performance

The dataset and literature reveal two geometrically distinct routes to simultaneously high RT and LT CE:

Route 1 — SCE* band (this work): Approach $\text{SCE}^* = 1.466$ via solvent mixture design. Mix molecular classes with different ESP profiles to create a coordination geometry distribution centred at the fixed point. Representative confirmed examples: LiFSI 1M DME/TMP; LiFSI 1M FEME/DME.

Route 2 — temperature-insensitive AGG: High salt concentration forces AGG-dominant solvation in which the anion is the primary coordinating species. AGG desolvation is temperature-insensitive because the barrier is set by anion- Li^+ binding (electrostatic, not geometry-dependent). Representative confirmed examples: LiFSI 4M DME; LiFSI 4M DME + BTfE (LHCE). Cost: viscosity increase, limited ionic conductivity.

These two routes are geometrically distinct and produce different failure modes under perturbation. Both are confirmed in the literature. $\text{SCE}^* = 1.466$ addresses Route 1 exclusively.

5.3 DEC Melting Point Correction

A systematic error in the published Li metal battery electrolyte literature was identified and is placed on record here.

Diethyl carbonate (DEC) is commonly listed as having a melting point of -43°C in electrolyte design literature. The primary CRC Handbook of Chemistry and Physics [17] and original characterisation data report $T_{\text{m}}(\text{DEC}) = -74.3^\circ\text{C}$. The figure -43°C appears to derive from a transcription or propagation error in secondary sources, repeated since approximately 1921.

Remark 5.1. This correction reclassifies EC/DEC-based electrolytes as liquids at temperatures previously considered problematic. It does not materially affect the SCE framework results (EC/DEC systems are R3 and excluded from the main gradient analysis), but it is placed on record as a contribution to the field’s reference data.

6 Forward Predictions and Candidate Formulations

The following candidate electrolytes are derived from the confirmed framework equations as targeting $\text{SCE}^* = 1.466$. All six are pre-registered before experiment in the public repository <https://github.com/Eric-Robert-Lawson/attractor-oncology>; the timestamped commit history is the priority record.

6.1 Structural Principle: Two Levers for Reaching SCE*

From below SCE^* (shell too ordered—add diversity): Mix molecular classes with different electrostatic potential (ESP) surface profiles. Cross-class mixing (e.g., linear ether + cyclic ether,

ether + phosphate ester) maximises structural tension in the coordination shell, shifting the population distribution toward SCE*.

From above SCE* (shell too disordered—remove diversity): Add an anion receptor (e.g., trimethyl borate, TMB) to reduce free anion coordination without altering solvent geometry, tightening the population distribution toward SCE*.

6.2 The Six Candidates

Candidate 1 — Cross-class linear/cyclic ether mixing **Formulation:** LiFSI 1.2 M in 2-MeTHF/DME (1.6:1 v/v)

Target SCE: 1.47 (estimated from ESP-weighted mixing)

Structural mechanism: 2-MeTHF (cyclic, SCE = 1.55) + DME (linear, SCE = 1.27) are structurally distinct classes. Mixed at 1.6:1 the shell populations bracket SCE* from both sides.

Falsification condition: $CE_{RT} > 89\%$ AND $CE_{LT} > 70\%$ at $-20\text{ }^{\circ}\text{C}$ (100 cycles, Cu/Li half cell). If $CE_{RT} < 85\%$ OR $CE_{LT} < 60\%$: framework prediction for this formulation is falsified.

Candidate 2 — Fluorinated/cyclic class mixing **Formulation:** LiFSI 1.0 M in FEME/2-MeTHF (60:40 v/v)

Target SCE: 1.47 (FEME at SCE = 1.37, 2-MeTHF at SCE = 1.55; 60:40 interpolates SCE*)

Structural mechanism: FEME fluorination reduces electron density at the ether oxygen, altering Li^+ -solvent binding geometry relative to non-fluorinated analogues. Mixed with 2-MeTHF the shell sees structurally distinct coordination modes.

Falsification condition: $CE_{RT} > 89\%$ AND $CE_{LT} > 72\%$ at $-20\text{ }^{\circ}\text{C}$.

Candidate 3 — Anion receptor from above SCE* **Formulation:** LiFSI 1.0 M in DOL/TMB (4:1 v/v)

Target SCE: 1.47 (DOL baseline SCE = 1.41; TMB anion receptor reduces free FSI^- coordination, shifting the population distribution upward toward SCE* from the ordered side)

Structural mechanism: TMB coordinates FSI^- preferentially, reducing anion participation in the solvation shell without altering solvent geometry directly. This removes a source of ordered (CIP/AGG) configurations, raising SCE toward SCE* from below.

Falsification condition: $CE_{RT} > 90\%$ AND $CE_{LT} > 70\%$ at $-20\text{ }^{\circ}\text{C}$.

Candidate 4 — Phosphate ester cross-class mixing **Formulation:** LiFSI 1.0–1.2 M in DME/TMP (variable ratio, target SCE = 1.47)

Target SCE: 1.47 (TMP is phosphate ester class; DME is linear ether class; the ESP profiles are effectively matched within 0.01 eV, maximising population splitting)

Structural mechanism: TMP (trimethyl phosphate) coordinates Li^+ via the phosphoryl oxygen with different geometry than DME ether oxygens. Matched ESP minima ($\Delta\text{ESP} = 0.007\text{ eV}$) prevent electrostatically preferential coordination of either species, splitting the population across both classes.

Falsification condition: $CE_{RT} > 88\%$ AND $CE_{LT} > 70\%$ at $-20\text{ }^{\circ}\text{C}$.

Candidate 5 — Ternary double cross-class **Formulation:** LiFSI 1.0 M in 2-MeTHF/FEME/DME (ternary, ratios TBD by SCE measurement)

Target SCE: 1.47 (three-component system with two ESP contrast axes)

Structural mechanism: Ternary cross-class mixing provides the maximum structural diversity

at minimum component count. Two distinct class contrasts simultaneously (linear/cyclic and fluorinated/non-fluorinated).

Falsification condition: $CE_{RT} > 90\%$ AND $CE_{LT} > 74\%$ at -20°C .

Candidate Arctic A — Extreme low temperature target **Formulation:** LiFSI 1.0 M in DME/TMP/TMB (three-class architecture, target $SCE = 2.0\text{--}2.5$)

Target SCE: Above SCE^* by design — optimising LT at the cost of RT. Three coordination classes (ether oxygen, phosphoryl oxygen, anion receptor) for maximum shell diversity.

Performance target: $CE_{LT} \geq 88\%$ at -40°C (Arctic operating temperature).

Falsification condition: $CE_{LT} < 75\%$ at -40°C : three-class architecture insufficient for arctic regime.

6.3 Experimental Protocol for Validation

For each candidate:

1. Measure coordination geometry populations by MD simulation or Raman spectroscopy; compute SCE from Eq. (1).
2. Confirm $SCE \in [1.40, 1.55]$ (within-band of SCE^*) for Candidates 1–5, or $SCE \in [2.0, 2.5]$ for Arctic A.
3. Measure CE_{RT} at 25°C and CE_{LT} at -20°C (or -40°C for Arctic A), 100 cycles, Cu/Li half cell, 0.5 mA cm^{-2} .
4. Apply falsification condition.

If SCE deviates from target by > 0.15 units: adjust formulation ratio and repeat Step 1 before proceeding to CE measurement.

7 Geometric Closure of the Search Space

A systematic molecular search (MU Molecular Universe protocol, full record at <https://github.com/Eric-Robert-Lawson/attractor-oncology>, file MU_Search_Complete_Space_Closure.md) was conducted to enumerate the available chemical space surrounding $SCE^* = 1.466$ using commercially available, non-exotic solvents.

7.1 Search Protocol and Results

Nine anchor searches were completed using SMILES-based molecular neighbourhood exploration. The search covered three chemical regions:

Region 1 — Ether class: Linear glymes (DME, DGM), cyclic ethers (DOL, 2-MeTHF, THF, THP), fluorinated linear ethers (FEME, TFME), fluorinated cyclic ethers. TMP consistently returned as the cross-class bridge anchor from all ether-anchored searches.

Region 2 — Phosphate ester class: TMP, TEP, TBP, and analogues. Confirmed as the universal cross-class attractor from ether-anchored searches; ESP contrast with ethers is maximised.

Region 3 — Boron ester class: TMB and analogues. Confirmed as the anion receptor class; reduces free anion coordination without altering solvent geometry.

Region 4 — Discards: Sulfolane and analogues returned from search but were excluded due to high viscosity at LT ($T_m = 27.5^\circ\text{C}$) and known incompatibility with Li metal at standard potentials.

7.2 Geometric Closure Statement

After nine anchor searches, the molecular space accessible from commercially available solvents is exhausted. Every molecular neighbourhood accessible from ether, sulfolane, amide, and phosphate anchors returns the same three regions. No additional distinct chemical class was identified.

Proposition 7.1 (Geometric closure). *The dual-temperature optimal electrolyte space for LiFSI-based systems, restricted to commercially available non-exotic solvents, is geometrically closed. It contains six named candidate formulations. No additional candidates are derivable from the accessible molecular universe without exotic synthesis.*

This closure converts an unbounded experimental search into a bounded set of six experiments. Fifty years of empirical screening is replaced by six named falsification tests.

8 Literature Position

8.1 Zero Incompatibilities Found

A systematic literature search across ten query families (full record at <https://github.com/Eric-Robert-Lawson/attractor-oncology>, file `Literature_Review_Compatibility_Check.md`) found zero papers containing claims geometrically incompatible with the SCE framework and the derived fixed point.

Every adjacent paper confirmed a facet of the framework’s structural predictions: solvation entropy as a performance variable (Joule 2025 [15]), ESP-matching as a screening tool (Nature Communications 2025 [16]), thermodynamic fixed points in coordination chemistry (compatible), solvation structure evolution at LT (compatible), and the Holoubek DOL/DME dataset (the negative control confirmation [3]).

8.2 What No Prior Work Contains

- No prior work derives $\text{SCE}^* = 1.466$.
- No prior work identifies the Shannon entropy of the Li^+ coordination geometry population distribution as the primary dual-temperature performance variable.
- No prior work derives a pre-registered fixed point from calculus applied to the combined RT+LT performance surface.
- No prior work uses the fixed point as a design target to close the candidate search space to a named, bounded set of formulations.

The fixed point is not in any prior paper. The variable is independently confirmed. The analytical derivation is this work.

9 Discussion

9.1 What the Fixed Point Changes

Prior to this framework, electrolyte design for Li metal batteries operated by empirical screening: formulate, test, measure CE, adjust, repeat. Without a geometric target, the search space was unbounded. Fifty years of work have produced a large literature of tested systems but no design rule that points to a specific location in formulation space.

The fixed point $SCE^* = 1.466$ changes the problem:

- Step 1: Measure the coordination geometry population (MD or Raman).
- Step 2: Compute SCE from Eq. (1).
- Step 3: If $SCE < SCE^*$: add structural diversity (cross-class ESP mixing, lower salt concentration). If $SCE > SCE^*$: reduce diversity (anion receptor, higher salt concentration).
- Step 4: Iterate to target. Test CE.

This is navigation with a map, not trial and error.

9.2 Generalisability

The framework is not Li-specific. SCE is defined for any metal ion with a measurable coordination geometry population distribution. The independent confirmation by Luo *et al.* (Joule 2025) demonstrating the variable operates for Na, K, Zn, and Al [15] confirms that the method generalises. Each metal ion chemistry has a fixed point derivable by the same calculus applied to a metal-ion-specific dataset. Those fixed points (SCE_{Na}^* , SCE_{Zn}^* , etc.) are named open derivations.

9.3 Limitations

- The confirmed equations are empirical fits to published data. The fixed point is analytically derived from those fits. The quality of the fixed point is bounded by the quality of the underlying dataset.
- The 21-system dataset is limited by the availability of systems for which explicit coordination geometry populations have been published. Additional data points will sharpen the confirmed equations.
- The candidate formulations are predicted, not confirmed. Experimental validation against the stated falsification conditions is the mandatory next step.
- Arctic Candidate A operates above SCE^* by design. Its LT performance prediction is a forward extrapolation of the band model beyond its confirmed domain ($SCE > 2.28$).
- The R3 (carbonate) regime is excluded from the gradient analysis. SCE may not be the primary performance variable for carbonate-class electrolytes, which form SEI through a different mechanism.

9.4 The Foundational Reframe

The Li metal dual-temperature electrolyte problem was not solved for fifty years despite enormous research effort. This work demonstrates that the solution was derivable from calculus applied to a small dataset of published coordination geometry populations.

The question that leads to the fixed point—"what Shannon entropy value simultaneously maximises both RT and LT CE?"—was not on the list of questions the domain framework generated. Not because the experts were incapable, but because the filter suppressed the question.

Remove the filter. The question becomes visible. The fixed point follows. That compression—from fifty years to a bounded derivation—is the deepest result of this work. The fixed point is the output. The method is the contribution.

10 Conclusion

We have derived the universal performance fixed point for dual-temperature Li metal battery electrolyte design:

$$\text{SCE}^* = 1.466 \quad (13)$$

This is the Shannon Coordination Entropy of the Li^+ first-shell solvation geometry population at the unique global maximum of combined room-temperature and low-temperature Coulombic efficiency. It is derived analytically, confirmed empirically across 21 published systems ($R^2 = 0.828$, $p = 4.5 \times 10^{-6}$), confirmed by negative control, and independently replicated across four metal ion chemistries.

Six candidate electrolyte formulations are derived and pre-registered with exact falsification conditions in the public repository at <https://github.com/Eric-Robert-Lawson/attractor-oncology>. The search space is geometrically closed from all accessible molecular directions. The experimental program is bounded.

The method is general. Every metal ion battery chemistry with a measurable coordination geometry population has a fixed point derivable by this framework. The derivation of those fixed points is the forward research program.

Data and code availability. The complete derivation record, dataset, R^2 progression (Steps 1–8), molecular search record, candidate derivations, and pre-registration record are archived in the public repository with full timestamped commit history: <https://github.com/Eric-Robert-Lawson/attractor-oncology>. The repository commit history constitutes the priority record for all predictions and falsification conditions stated in this paper.

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Author contributions. E.R.L.: conceptualisation, derivation, analysis, writing.

A Derivation of SCE^* from the Confirmed Equations

This appendix presents the full algebraic derivation for transparency.

Inputs:

$$\begin{aligned} f_1(\text{SCE}) &= 100.1 - e^{1.493 + 1.230 \cdot \text{SCE}} \quad (\text{CE}_{\text{RT}}, \text{Regime 1}) \\ f_2(\text{SCE}) &= 11.91 + 33.21 \cdot \text{SCE} \quad (\text{CE}_{\text{LT}}, \text{band}) \end{aligned}$$

First-order condition:

$$\begin{aligned}
\frac{d}{d(\text{SCE})} [f_1 + f_2] &= 0 \\
-1.230 \cdot e^{1.493+1.230 \cdot \text{SCE}} + 33.21 &= 0 \\
e^{1.493+1.230 \cdot \text{SCE}^*} &= \frac{33.21}{1.230} = 27.0 \\
1.493 + 1.230 \cdot \text{SCE}^* &= \ln(27.0) = 3.296 \\
\text{SCE}^* &= \frac{3.296 - 1.493}{1.230} = \frac{1.803}{1.230} = 1.466
\end{aligned}$$

Second derivative:

$$\begin{aligned}
\frac{d^2}{d(\text{SCE})^2} [f_1 + f_2] \Big|_{\text{SCE}^*} &= -(1.230)^2 \cdot e^{1.493+1.230 \times 1.466} \\
&= -1.5129 \times 27.0 = -40.85 < 0 \quad \checkmark
\end{aligned}$$

The fixed point is a unique global maximum.

B Confirmed Parameter Values and Confidence Intervals

Table 3: Confirmed regression parameters for the four framework equations. CI = 95% confidence interval. Full numerical output available in `step8_findings_report.json` in the repository.

Equation	b_0	b_1	SE(b_1)	t	p	95% CI(b_1)
Eq. 1 (RT deficit, R1)	1.493	1.230	0.322	3.82	0.009	[0.441, 2.018]
Eq. 2 (LT band)	11.91	33.21	—	—	0.025	—
Eq. 3 (two-variable, R1+R2)	104.68	-25.85	—	—	4.5×10^{-6}	—

Note: Equation 3 includes a regime dummy variable (coefficient +31.16 for Regime 2 membership) alongside the SCE term. Full model output is in the repository.

C Pre-Registered Falsification Conditions

The following conditions were committed to the public repository at <https://github.com/Eric-Robert-Lawson/attractor-oncology> before any experimental measurement of the candidate formulations was performed. The repository commit history is the timestamped priority record.

1. **Candidate 1** (LiFSI 1.2M, 2-MeTHF/DME 1.6:1 v/v):
Framework falsified if $\text{CE}_{\text{RT}} < 85\%$ OR $\text{CE}_{\text{LT}} < 60\%$.
2. **Candidate 2** (LiFSI 1.0M, FEME/2-MeTHF 60:40 v/v):
Framework falsified if $\text{CE}_{\text{RT}} < 85\%$ OR $\text{CE}_{\text{LT}} < 62\%$.
3. **Candidate 3** (LiFSI 1.0M, DOL/TMB 4:1 v/v):
Framework falsified if $\text{CE}_{\text{RT}} < 87\%$ OR $\text{CE}_{\text{LT}} < 60\%$.
4. **Candidate 4** (LiFSI 1.0–1.2M, DME/TMP variable ratio):
Framework falsified if $\text{CE}_{\text{RT}} < 84\%$ OR $\text{CE}_{\text{LT}} < 60\%$.

5. **Candidate 5** (LiFSI 1.0M, 2-MeTHF/FEME/DME ternary):
Framework falsified if $CE_{RT} < 87\%$ OR $CE_{LT} < 64\%$.
6. **Arctic Candidate A** (LiFSI 1.0M, DME/TMP/TMB 1:1:1 starting ratio):
Framework falsified if $CE_{LT}(-40\text{ }^{\circ}\text{C}) < 75\%$.

D DEC Melting Point Primary Source Record

The melting point of diethyl carbonate (DEC) is commonly cited as $-43\text{ }^{\circ}\text{C}$ in the electrolyte design literature.

Primary source: CRC Handbook of Chemistry and Physics, 105th edition [17], entry for diethyl carbonate (CAS 105-58-8): $T_m(\text{DEC}) = -74.3\text{ }^{\circ}\text{C}$.

The figure $-43\text{ }^{\circ}\text{C}$ does not appear in the primary source. Its origin in the secondary literature has not been traced to a specific experimental measurement. All future references to DEC in the context of low-temperature electrolyte design should use $-74.3\text{ }^{\circ}\text{C}$.

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